Collider Physics of the Inert Doublet model

Honours Project 2 Report

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1 Introduction

The Standard Model (SM) of particle physics, while remarkably successful, fails to account for approximately 27% of the energy density of the universe—the component known as Dark Matter (DM). The Inert Doublet Model (IDM) is one of the most compelling and minimal extensions of the SM that provides a viable Cold Dark Matter candidate. This report explores the collider phenomenology of the IDM, focusing on its production and detection at the Large Hadron Collider (LHC). The code and UFO files used can be found at github.

2 Inert Doublet Model

The Inert 2HDM is a minimalist extension of Standard Model. The Lagrangian is invariant under $\mathbb{Z}_2$ parity transformation where, $\Phi_1 \to \Phi_1$ and $\Phi_2 \to -\Phi_2$. The most general potential is given as,

$$\begin{aligned} V = {} & m_{11}^2 \Phi_1^\dagger \Phi_1 + m_{22}^2 \Phi_2^\dagger \Phi_2 + \lambda_1 (\Phi_1^\dagger \Phi_1)^2 + \lambda_2 (\Phi_2^\dagger \Phi_2)^2 \\ & + \lambda_3 (\Phi_1^\dagger \Phi_1)(\Phi_2^\dagger \Phi_2) + \lambda_4 (\Phi_1^\dagger \Phi_2)(\Phi_2^\dagger \Phi_1) \\ & + \left\{ \lambda_5 (\Phi_1^\dagger \Phi_2)^2 + \text{h.c.} \right\} \end{aligned} \tag{2}$$

where,

$$\Phi_1 = \begin{pmatrix} \phi_1^+ \\ \phi_1^0 \end{pmatrix}, \qquad \Phi_2 = \begin{pmatrix} \phi_2^+ \\ \phi_2^0 \end{pmatrix}$$

and m_{11}^2, m_{22}^2 and $\lambda_{1,2,3,4,5}$ are real parameters. After Electroweak Symmetry Breaking, Φ_2 does not get any vev as it is $\mathbb{Z}_2$-odd. Only Φ_1 gets a non-zero vev. At Electroweak minima,

$$\langle \Phi_1 \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}$$

with $v \simeq 246$ GeV. Now, using minimization condition, the mass parameter m_{11}^2 can be expressed in terms of other parameters as,

$$m_{11}^2 = -v^2 \lambda_1 \tag{3}$$

Now, writting the scalar fields as,

$$\langle \Phi_1 \rangle = \begin{pmatrix} \phi_1^+ \\ \frac{1}{\sqrt{2}}(v + h_1 + i n_1) \end{pmatrix}, \qquad \langle \Phi_2 \rangle = \begin{pmatrix} \phi_2^+ \\ \frac{1}{\sqrt{2}}(h_2 + i n_2) \end{pmatrix}$$

and expanding the potential, it is concluded that out of the eight degrees of freedom, three Goldstone Bosons ($G^\pm, G^0$) are absorbed by the W and Z bosons. The remaining five physical higgs particles are: two CP-even higgs h & H, one CP-odd higgs A, and one charged higgs pair $H^\pm$.

After the electroweak symmetry breaking, the masses of the scalar particles are given by:

$$M_h^2 = 2\lambda_1 v^2 \tag{4}$$

$$M_H^2 = m_{22}^2 + \frac{v^2}{2}(\lambda_3 + \lambda_4 + 2\lambda_5) = m_{22}^2 + \frac{v^2}{2}\lambda_L \tag{5}$$

$$M_A^2 = m_{22}^2 + \frac{v^2}{2}(\lambda_3 + \lambda_4 - 2\lambda_5) = m_{22}^2 + \frac{v^2}{2}\lambda_S \tag{6}$$

$$M_{H^\pm}^2 = m_{22}^2 + \frac{v^2}{2}\lambda_5 \tag{7}$$

3 Production Mechanisms and Cross-Section Analysis

This section details the cross-section (σ) behavior for various inert scalar production channels as a function of the mass scale.

3.1 Theoretical Production Channels

At a $\sqrt{s} = 13$ TeV proton-proton collider, the IDM scalars are produced primarily through electroweak interactions. The main production modes considered are:

- **Pair Production:** Charged scalar pairs (H^+H^-) and neutral-charged pairs ($A^0H^\pm$, $H^0H^\pm$).
- **Decay-Mediated Production:** Associated production involving $W^\pm$ bosons through Higgs decays such as $h \to W^\pm A^0$ and $h \to W^\pm H^0$.

3.2 Cross-Section Results

The plots below represent the results of a parameter scan performed across a mass range of 0 to 500 GeV. These values were calculated using the IDM model implementation in `MadGraph5` and plotted via python

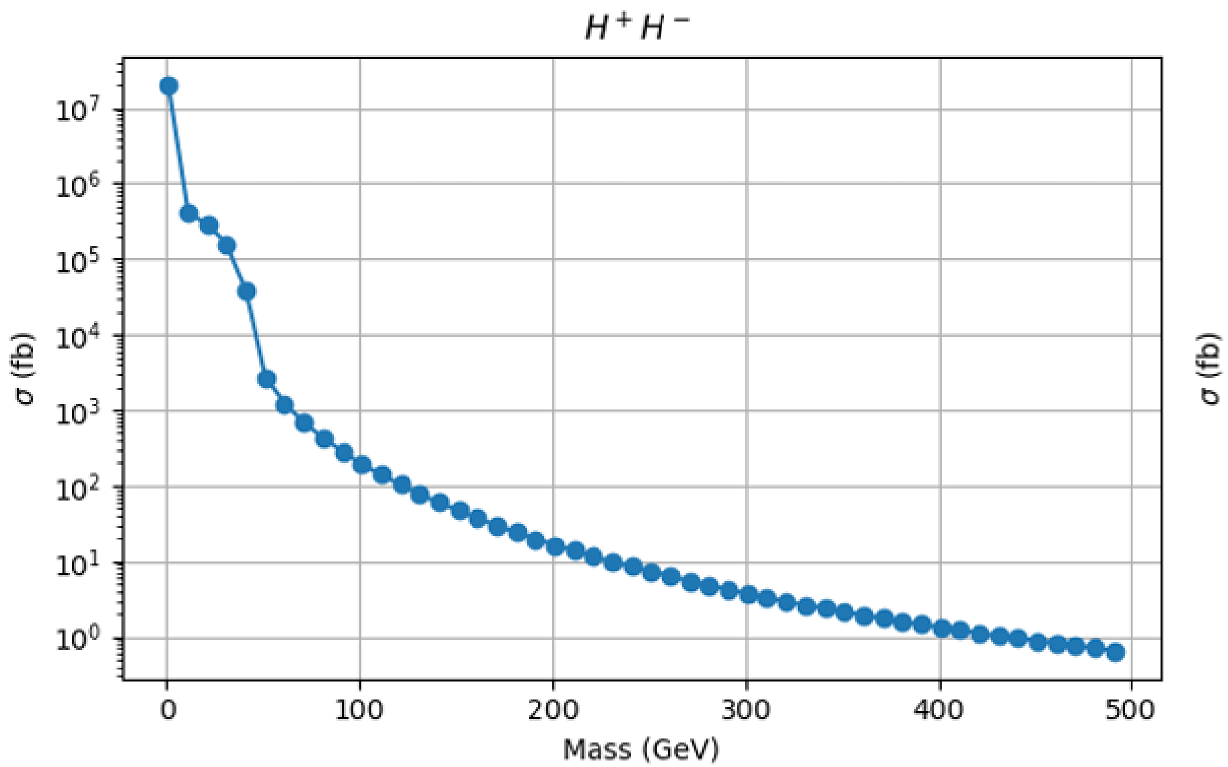


Figure 1: H^+H^- production cross-section as a function of mass.

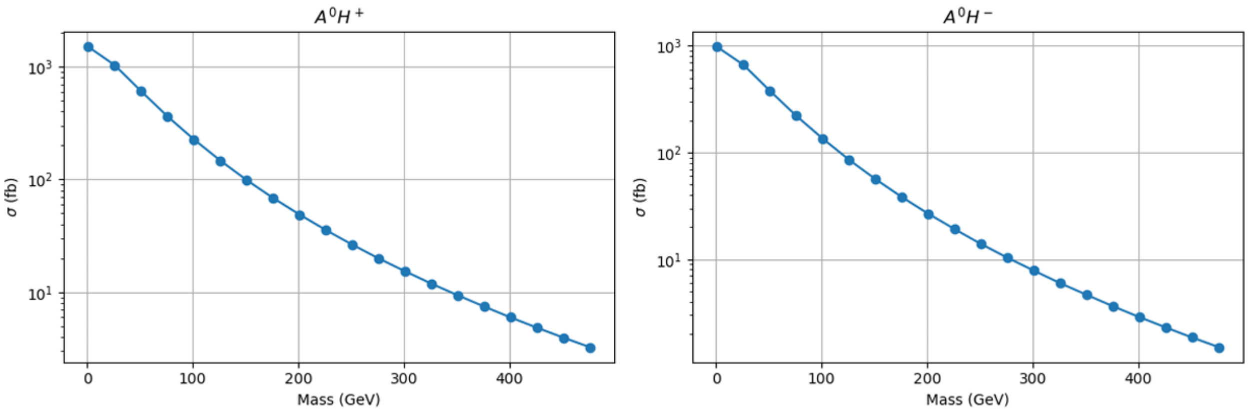


Figure 2: $A^0H^{\pm}$ production cross-section as a function of mass.

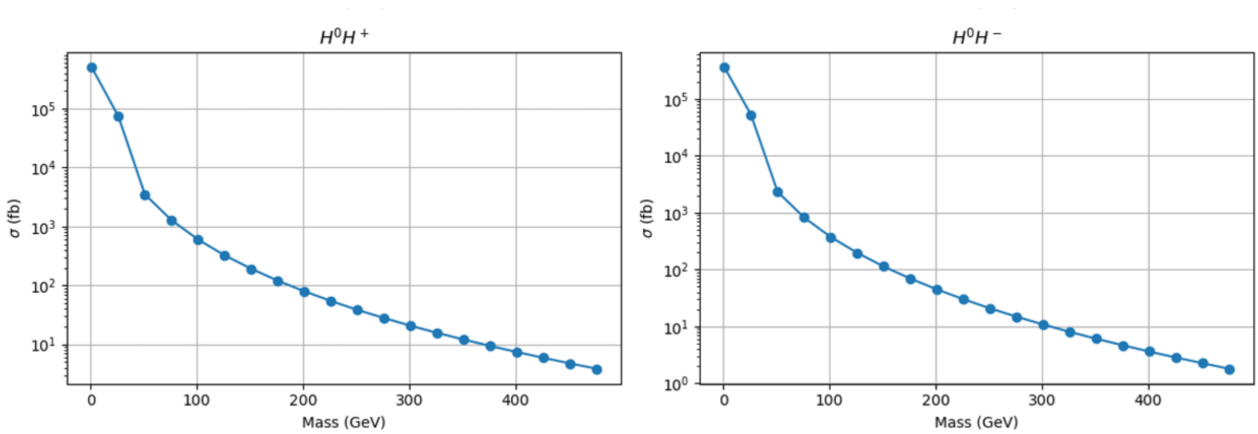
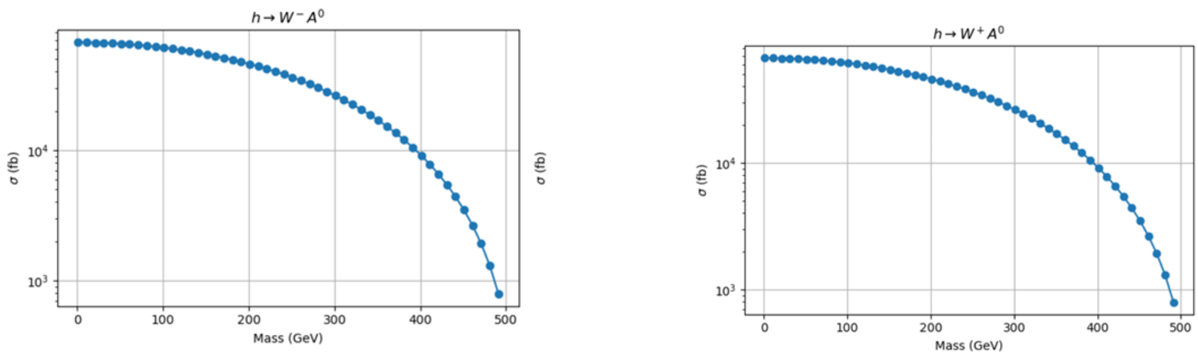


Figure 3: $H^0H^{\pm}$ production cross-section as a function of mass.



(a) $A^0H^{\pm}$ channel

(b) $H^0H^{\pm}$ channel

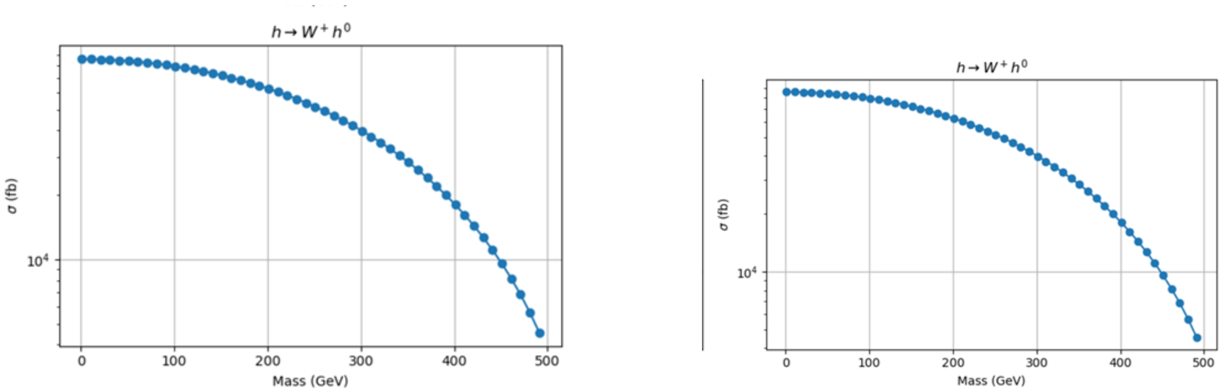


Figure 5: Cross-section scaling features across mass regimes.

Key Observations from the Data

- **Mass Dependence:** All channels exhibit a characteristic decrease in cross-section with increasing mass, consistent with phase-space suppression for heavy particle production.
- **Logarithmic Scaling:** The production rates span several orders of magnitude, ranging from $\mathcal{O}(10^4 \text{ fb})$ at low masses to below 1 fb near 500 GeV.
- **Channel Comparison:** Neutral-charged associated production channels ($A^0H^\pm$ and $H^0H^\pm$) generally yield higher production rates compared to pure charged pair production at higher mass scales, making them promising channels for experimental searches.

4 Further Study of the $H^\pm$ decay and significance calculation

In the Inert Doublet Model (IDM), the charged scalar $H^\pm$ predominantly decays into a W boson and the lightest neutral inert scalar H^0, which serves as the dark matter candidate. The decay process is given by:

$$H^\pm \to W^\pm + H^0 \tag{1}$$

The final state topology depends on the decay of the two W bosons (W^+W^-). Each W boson can decay either hadronically (into quarks) or leptonically (into a lepton and a neutrino).

Using the Standard Model branching fractions:

$$B_{\text{had}} \approx 0.67, \qquad B_{\text{lep}} \approx 0.33,$$

the probabilities for different final states are calculated as follows:

- **Fully Hadronic Channel ($\sim 45\%$):** Both W bosons decay hadronically:

$$P_{\text{had}} = B_{\text{had}} \times B_{\text{had}} \approx (0.67)^2 \approx 0.45 \tag{2}$$

- **Semi-Leptonic Channel ($\sim 44\%$):** One W decays hadronically and the other leptonically. There are two possible configurations:

$$P_{\text{semi}} = 2 \times (B_{\text{had}} \times B_{\text{lep}}) \approx 2 \times (0.67 \times 0.33) \approx 0.44 \tag{3}$$

- **Dileptonic Channel ($\sim 11\%$):** Both W bosons decay leptonically:

$$P_{\text{di}} = B_{\text{lep}} \times B_{\text{lep}} \approx (0.33)^2 \approx 0.11 \tag{4}$$

These ratios (45:44:11) provide a useful guide for categorizing signal regions in collider analyses, with each channel offering different advantages in terms of background suppression and signal sensitivity.

Since H^0 is stable and does not interact with the detector, it appears as missing transverse energy (MTE). Therefore, the observable signatures at the LHC are determined entirely by the decay modes of the $W^\pm$ bosons produced in H^+H^- pair production.

Table 1: Theoretical Branching Ratios for H^+H^- decay via W^+W^- final states.

Channel	**Decay Topology**	**Branching Ratio (%)**
Fully Hadronic	$q\bar{q}q\bar{q} + \text{MET}$	$\sim 45\%$
Semi-Leptonic	$\ell\nu q\bar{q} + \text{MET}$	$\sim 44\%$
Dileptonic	$\ell\nu\ell\nu + \text{MET}$	$\sim 11\%$

4.1 Significance calculation

Following the production of IDM scalars, the resulting W^+W^- system is analyzed across three distinct decay topologies: Hadronic, Semi-Leptonic, and Dileptonic. This section details the reconstruction of these final states and the subsequent calculation of the discovery potential at the LHC.

The simulated events were processed through Delphes to account for realistic detector effects. As established in our cross-section baseline, we assume an integrated luminosity of 137 fb^{-1}.

The primary background for this search is the Standard Model (SM) production of W^+W^- pairs. With a cross-section of 64.77 pb, the total expected background events at 137 fb^{-1} is calculated as:

$$B_{\text{total}} = \sigma \times \mathcal{L} \approx 8{,}873{,}490 \text{ events} \tag{5}$$

The statistical significance (Z) is calculated using the Gaussian approximation:

$$Z = \frac{S}{\sqrt{B}} \tag{6}$$

Selection Efficiency (ϵ) represents the probability that a generated signal event is successfully reconstructed by the detector and passes all specific kinematic and object-identification criteria for a given channel. The Signal Yield (S) is calculated using the following relationship:

$$S = \sigma \times \mathcal{L} \times \epsilon \tag{7}$$

In this formula, σ denotes the theoretical production cross-section, $\mathcal{L}$ is the integrated luminosity of the dataset, and ϵ is the selection efficiency derived from the simulated data.

The following table summarizes the baseline results obtained from our initial simulation run, assuming a theoretical signal cross-section of 500 fb:

Table 2: Reconstructed Signal Yields and Statistical Significance at $\mathcal{L} = 137$ fb^{-1}.

Channel	**Efficiency (ϵ)**	**Final Signal (S)**	**Significance (Z)**
IDM_Hadronic	99.94%	68,458.9	22.98 σ
IDM_SemiLeptonic	63.17%	43,271.5	14.53 σ
IDM_Dileptonic	44.36%	30,386.6	10.20 σ

5 Conclusion

In this work, we have explored the collider phenomenology of the Inert Doublet Model (IDM), a minimal and well-motivated extension of the Standard Model that provides a viable dark matter candidate. The structure of the model, characterized by a $\mathbb{Z}_2$ symmetry, ensures the stability of the lightest neutral scalar H^0, making it a suitable dark matter candidate.

The production mechanisms of inert scalars at the LHC were analyzed, with results showing that the cross-sections decrease significantly with increasing mass, consistent with phase-space suppression. Among the production channels, neutral-charged associated production ($A^0H^\pm$ and $H^0H^\pm$) remains relatively dominant at higher mass scales, indicating their importance for experimental searches.

The decay of the charged scalar $H^\pm \to W^\pm H^0$ was studied in detail, and the resulting final states were categorized into fully hadronic, semi-leptonic, and dileptonic channels. The relative contributions of these channels follow the expected Standard Model W boson branching fractions, yielding the characteristic 45:44:11 ratio.

A preliminary detector-level analysis was performed using Delphes. The statistical significance was estimated using the Gaussian approximation, with results indicating that the hadronic and semi-leptonic channels provide the strongest sensitivity, while the dileptonic channel, although cleaner, is statistically limited.

Overall, this study demonstrates that the IDM provides promising collider signatures at the LHC, particularly in channels involving missing transverse energy and W boson decay products. Future work can focus on improving signal extraction through optimized selection cuts, background reduction strategies, and more precise detector simulations to enhance the discovery potential of inert scalars.

References

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[2] A. Arhrib, R. Benbrik, M. El Kacimi, L. Rahili and S. Moretti, "The Inert Doublet Model: Constraints and Phenomenology," arXiv:1307.6346 [hep-ph].